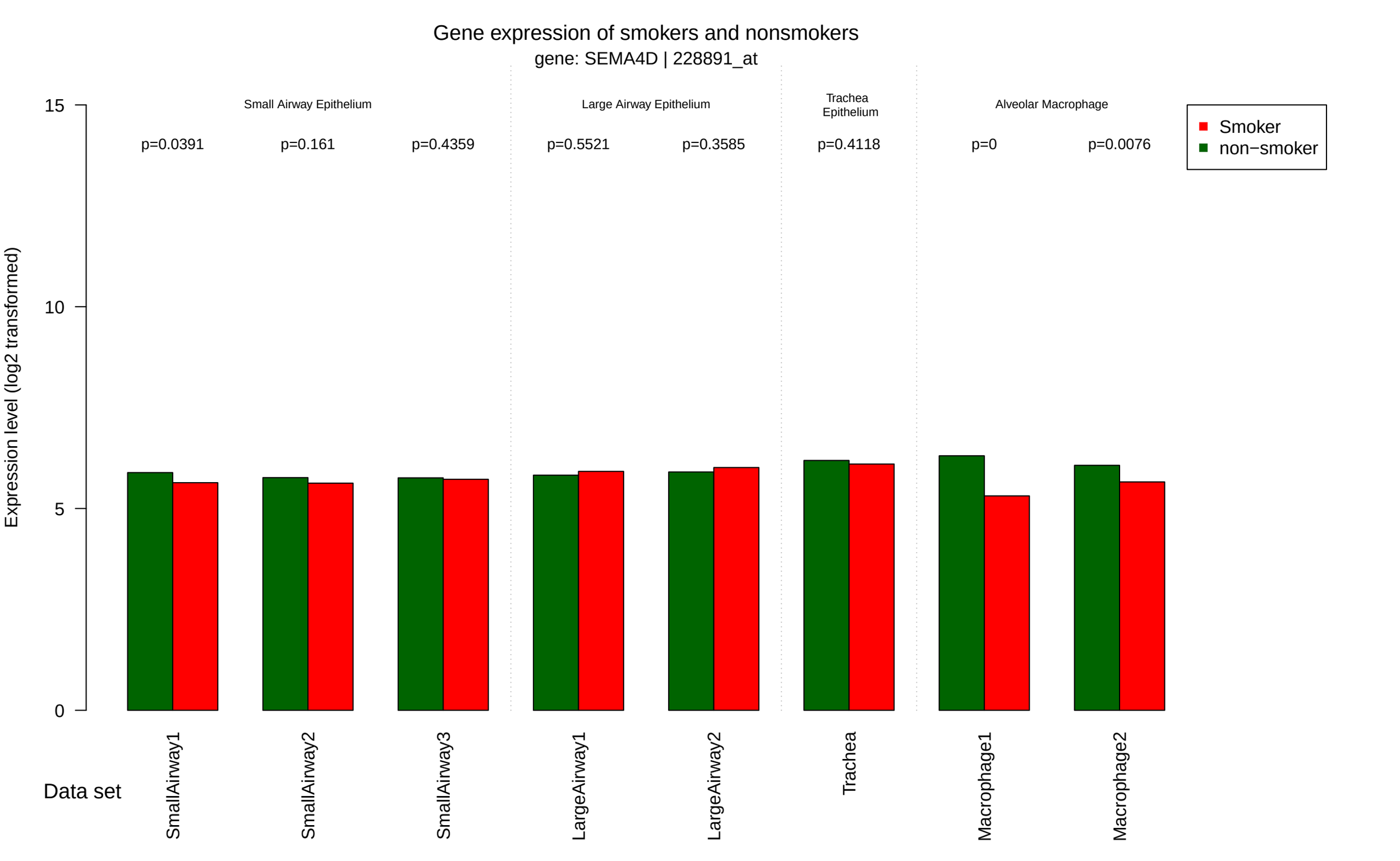
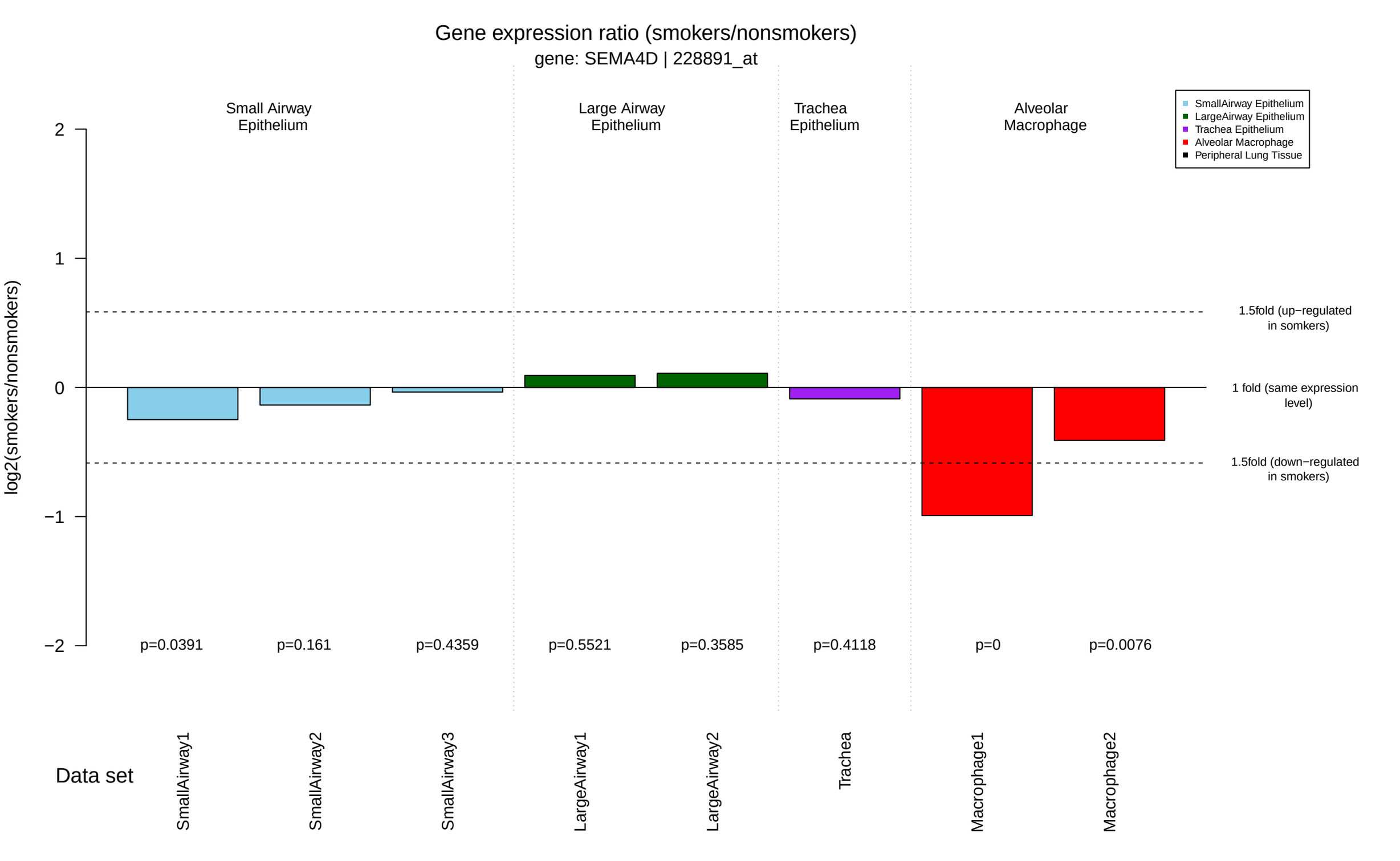
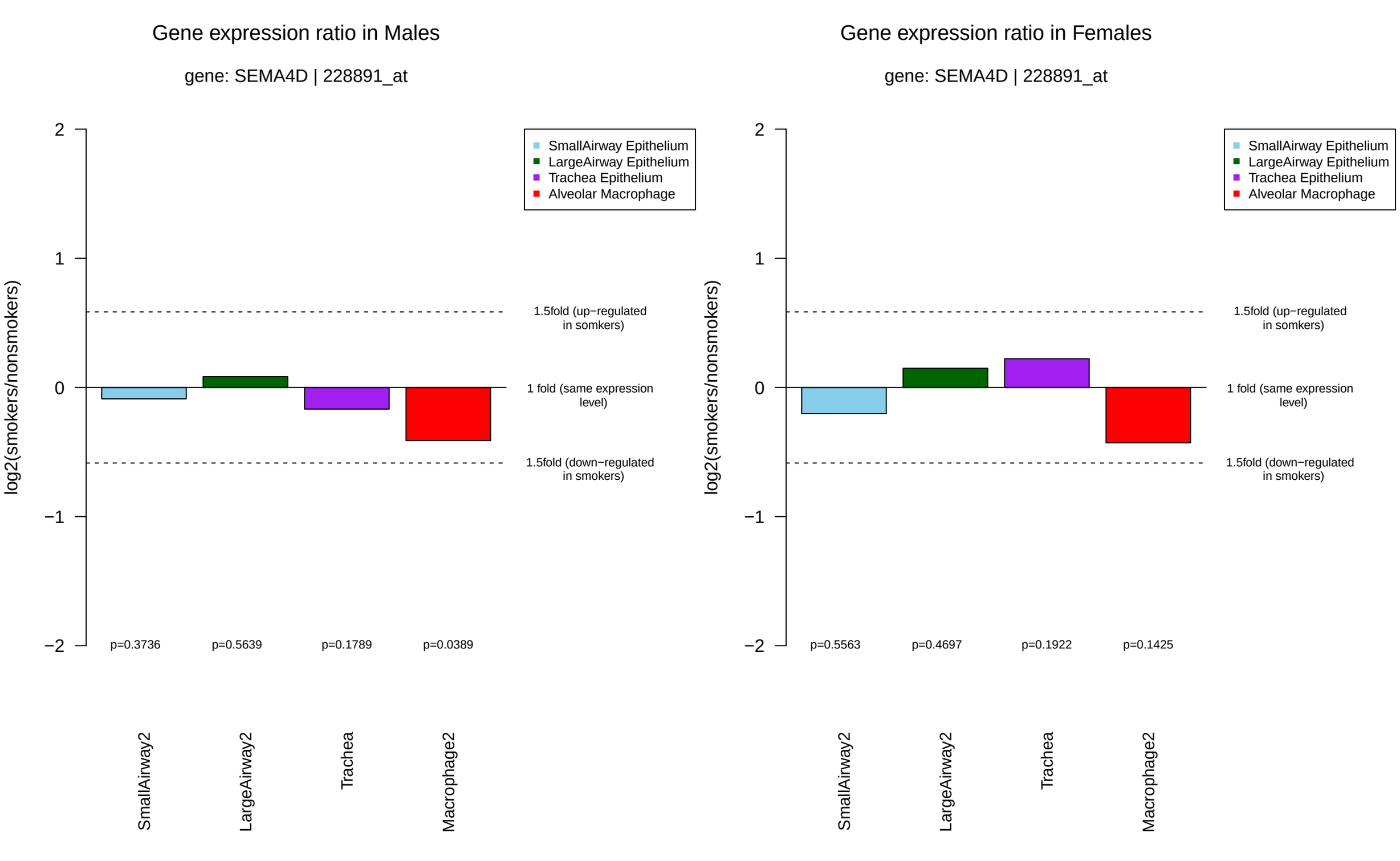
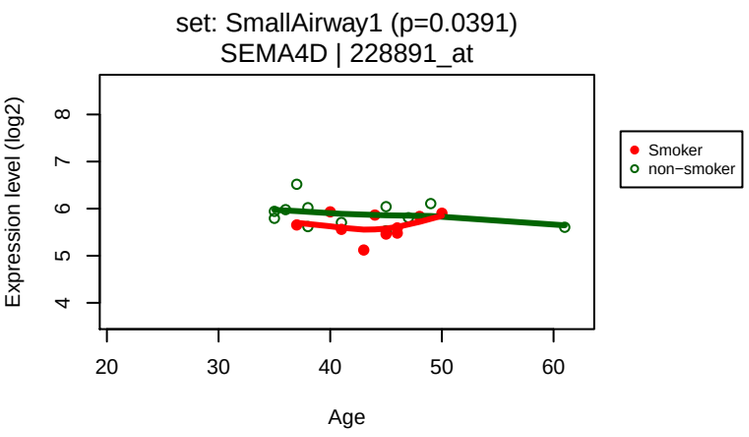
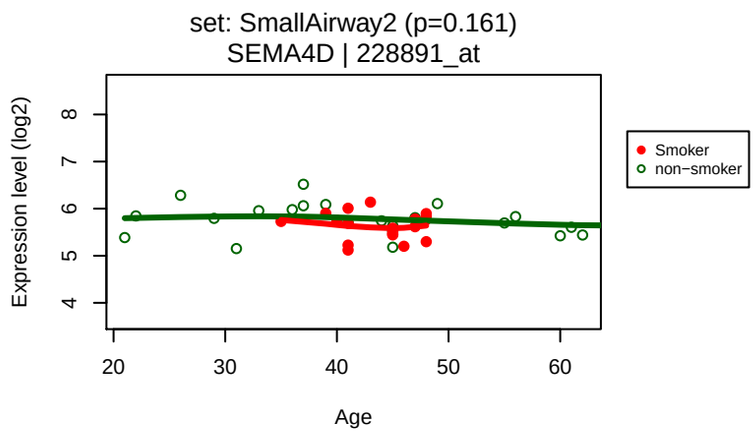
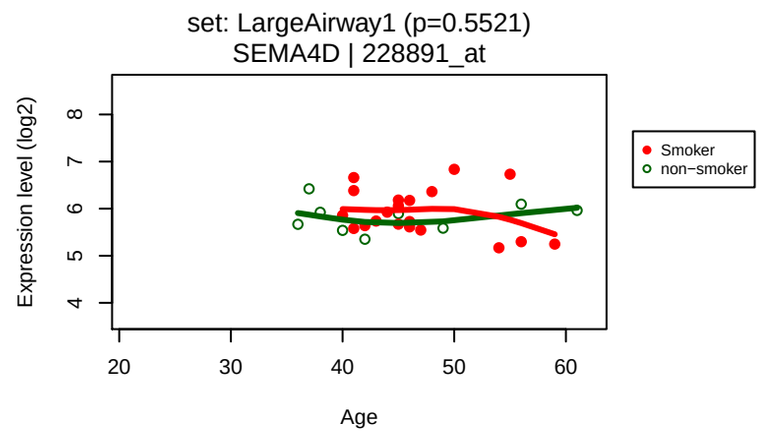

Gene expression of smokers and nonsmokers
gene: SEMA4D | 228891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0391
p=0.161
p=0.4359
p=0.5521
p=0.3585
p=0.4118
p=0
p=0.0076
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

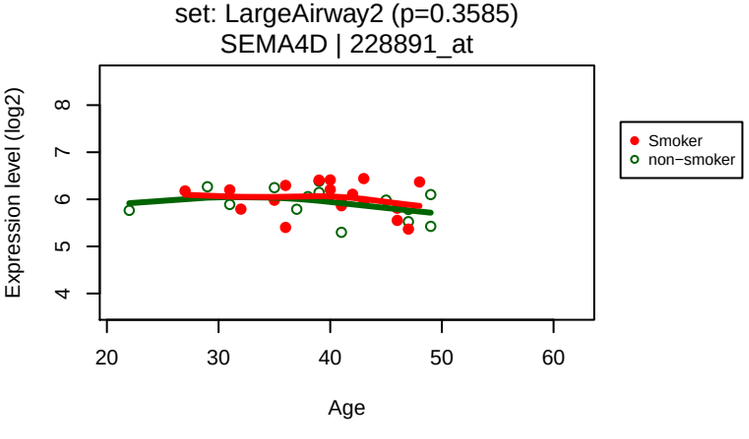

Gene expression ratio (smokers/nonsmokers)
gene: SEMA4D | 228891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0391
p=0.161
p=0.4359
p=0.5521
p=0.3585
p=0.4118
p=0
p=0.0076
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

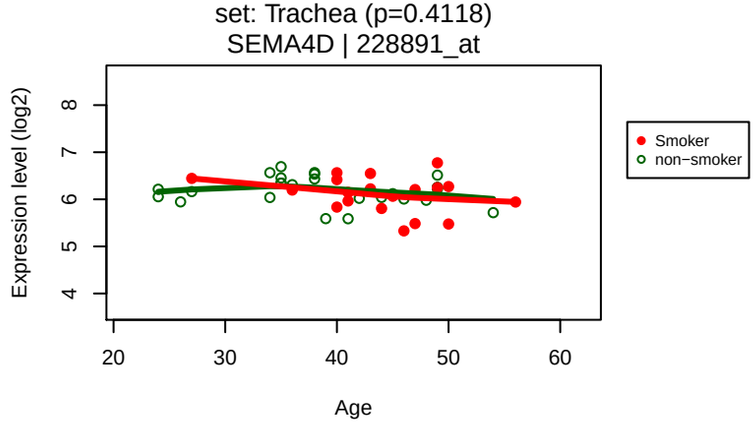

Gene expression ratio in Males
gene: SEMA4D | 228891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3736
p=0.5639
p=0.1789
p=0.0389
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA4D | 228891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5563
p=0.4697
p=0.1922
p=0.1425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

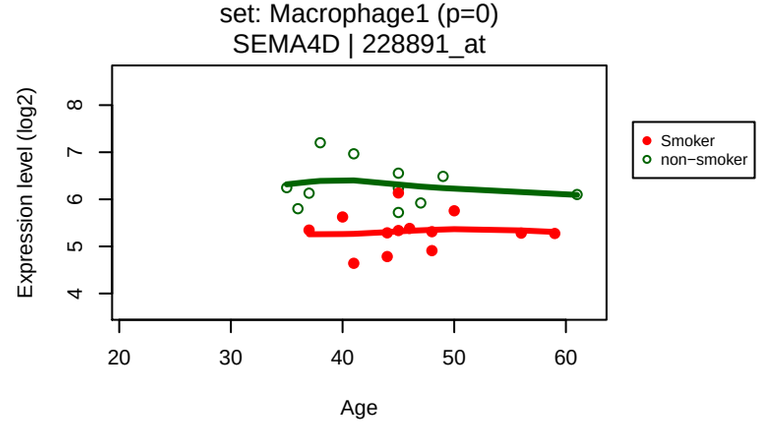

set: SmallAirway1 (p=0.0391)
SEMA4D | 228891_at
Expression level (log2)
Age
Smoker
non−smoker

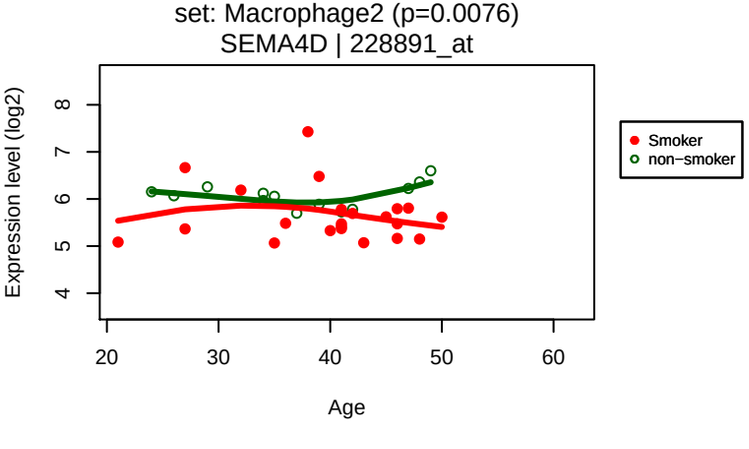

set: SmallAirway2 (p=0.161)
SEMA4D | 228891_at
Expression level (log2)
Age
Smoker
non−smoker

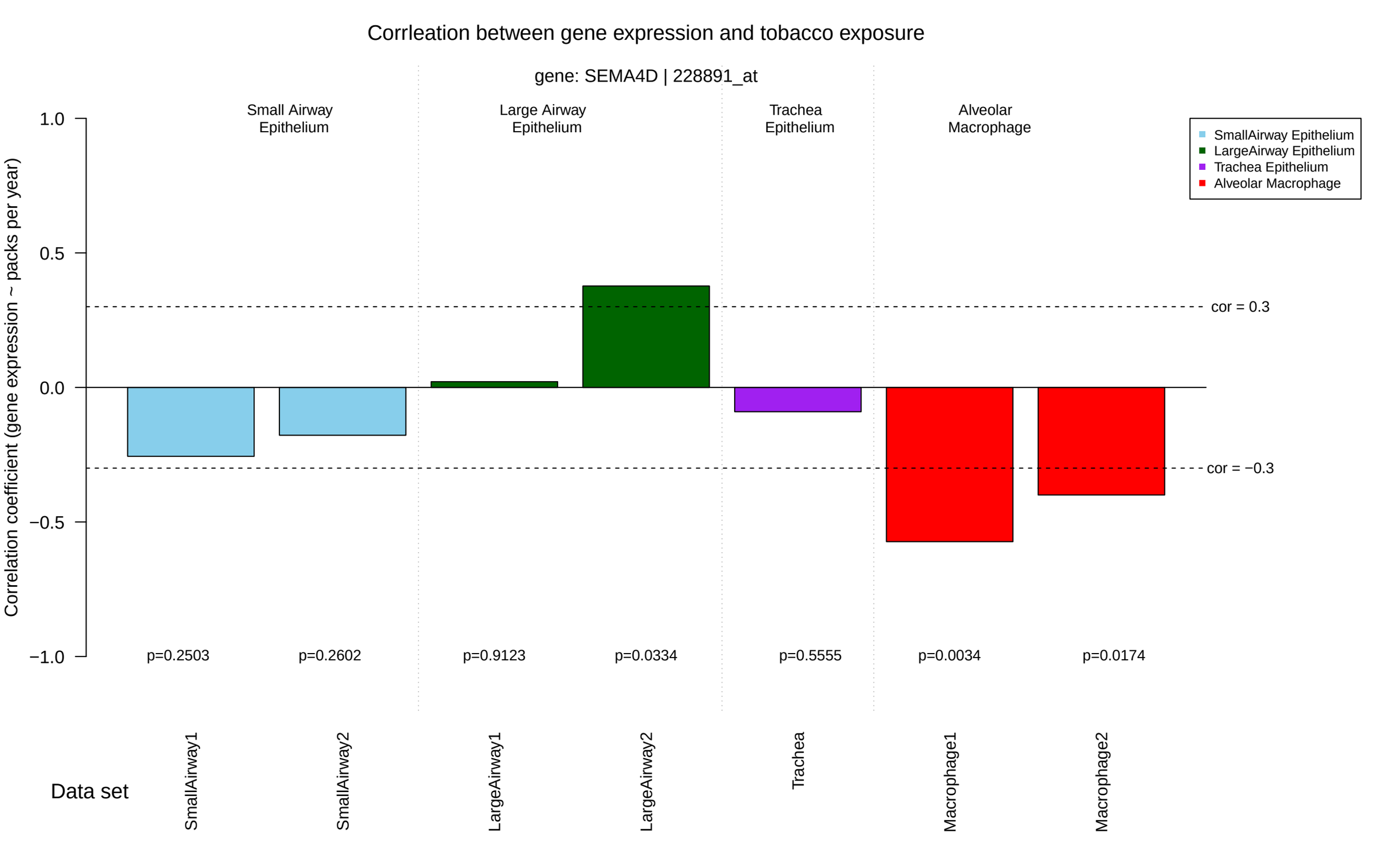

set: LargeAirway1 (p=0.5521)
SEMA4D | 228891_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3585)
SEMA4D | 228891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4118)
SEMA4D | 228891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0)
SEMA4D | 228891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0076)
SEMA4D | 228891_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA4D | 228891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2503
p=0.2602
p=0.9123
p=0.0334
p=0.5555
p=0.0034
p=0.0174
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2